Part A :Unveiling Customer Insights in the Auto Parts Manufacturing company

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Module: Marketing and Business Analysis

Executive Summary

- The dataset, consisting of 2747 entries and 20 columns, reveals comprehensive order and customer information, with essential features like ORDER NUMBER, QUANTITY ORDERED, and SALES.
- A meticulous exploration of numerical and categorical features, along with insightful statistics, uncovers notable trends, variability, and key quartile values, providing a comprehensive understanding of the dataset's characteristics.
- Sales insights showcase top-performing countries, products, and cities, emphasizing the dominance of USA, Spain, and France in both value and unit quantity, while specific product categories and geographical locations exhibit significant sales contributions.
- Detailed analyses highlight notable clients such as Euro Shopping Channel, Mini Gifts Distributors Ltd., and Australian Collectors, Co., along with RFM analysis categorizing customers based on recency, frequency, and monetary value.
- The analysis concludes with a focus on customer retention strategies, pinpointing those on the verge of churning and lost customers, providing actionable insights to enhance overall business performance.

Contents

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- Business Overview
- Exploratory data analysis
- Customer segmentation using RFM technique and Outputs
- KNIME workflow diagram
- Inferences from RFM Analysis

Problem Statement

- The automobile parts manufacturing company has accumulated three years' raw transactional data without the means to extract valuable insights.
- The objective is to unveil the purchasing patterns shaping their customer interactions by exploring the data and using RFM techniques.
- Our focus is to understand customer behaviors and crafting customized marketing strategies tailored to distinct customer segments.
- Through this analysis we bridge the gap between data and strategy, empower the client with actionable insights that drive to make informed decision-making and foster strategic marketing initiatives tailored to their diverse customer base in a dynamic industry.

Summary about the Data

Data type description :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2747 entries, 0 to 2746
Data columns (total 20 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   ORDERNUMBER           2747 non-null   int64
 1   QUANTITYORDERED       2747 non-null   int64
 2   PRICEEACH             2747 non-null   float64
 3   ORDERLINENUMBER       2747 non-null   int64
 4   SALES                 2747 non-null   float64
 5   ORDERDATE             2747 non-null   int64
 6   DAYS_SINCE_LASTORDER  2747 non-null   int64
 7   STATUS                2747 non-null   object
 8   PRODUCTLINE           2747 non-null   object
 9   MSRP                  2747 non-null   int64
 10  PRODUCTCODE           2747 non-null   object
 11  CUSTOMERNAME          2747 non-null   object
 12  PHONE                 2747 non-null   object
 13  ADDRESSLINE1          2747 non-null   object
 14  CITY                  2747 non-null   object
 15  POSTALCODE            2747 non-null   object
 16  COUNTRY               2747 non-null   object
 17  CONTACTLASTNAME       2747 non-null   object
 18  CONTACTFIRSTNAME      2747 non-null   object
 19  DEALSIZE              2747 non-null   object
dtypes: float64(2), int64(6), object(12)
```

- The dataset comprises **2747** entries with **20** columns, including order details and customer information.
- Essential **numerical features** include ‘ORDERNUMBER, QUANTITYORDERED, PRICEEACH, ORDERLINENUMBER, SALES, ORDER DATE, and DAYS_SINCE_LASTORDER.
- **Categorical features** encompass STATUS, PRODUCTLINE, PRODUCTCODE, CUSTOMERNAME, PHONE, ADDRESSLINE1, CITY, POSTALCODE, COUNTRY,CONTACTLASTNAME, CONTACTFIRSTNAME, and DEALSIZE.
- **No missing values** are present, and data types include int64 (6), float64(2), and object(12)

Summary about the Data

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	ORDERDATE	DAYS_SINCE_LASTORDER	MSRP
count	2747.000000	2747.000000	2747.000000	2747.000000	2747.000000	2747.000000	2747.000000	2747.000000
mean	10259.761558	35.103021	101.098951	6.491081	3553.047583	43598.914088	1757.085912	100.691664
std	91.877521	9.762135	42.042548	4.230544	1838.953901	230.231295	819.280576	40.114802
min	10100.000000	6.000000	26.880000	1.000000	482.130000	43106.000000	42.000000	33.000000
25%	10181.000000	27.000000	68.745000	3.000000	2204.350000	43412.000000	1077.000000	68.000000
50%	10264.000000	35.000000	95.550000	6.000000	3184.800000	43640.000000	1761.000000	99.000000
75%	10334.500000	43.000000	127.100000	9.000000	4503.095000	43786.000000	2436.500000	124.000000
max	10425.000000	97.000000	252.870000	18.000000	14082.800000	43982.000000	3562.000000	214.000000

1. Average order characteristics include a mean order quantity of 35, a mean price of $101.10, and an average sales value of $3553.05.
2. Standard deviations highlight variability, particularly in SALES and DAYS_SINCE_LASTORDER, indicating fluctuations around their means.
3. Insights from minimum and maximum values: ORDERNUMBER starts from 10100, ORDERDATE has a minimum of 43106, and the highest SALES value is $14082.80.
4. Quartile values (25%, 50%, 75%) provide insights into the distribution and spread of key features like QUANTITYORDERED, PRICEEACH, SALES, ORDERDATE, DAYS_SINCE_LASTORDER, and MSRP.

Business Overview

Total Sales ($)

9,760,222

Total Orders

96,428

Total Unique Customers

89

Last Qtr Sales ($)

719,494

Last Qtr Orders

6,991

Total Countries

19

Total Cities

71

EDA - Country wise summary - Auto Parts

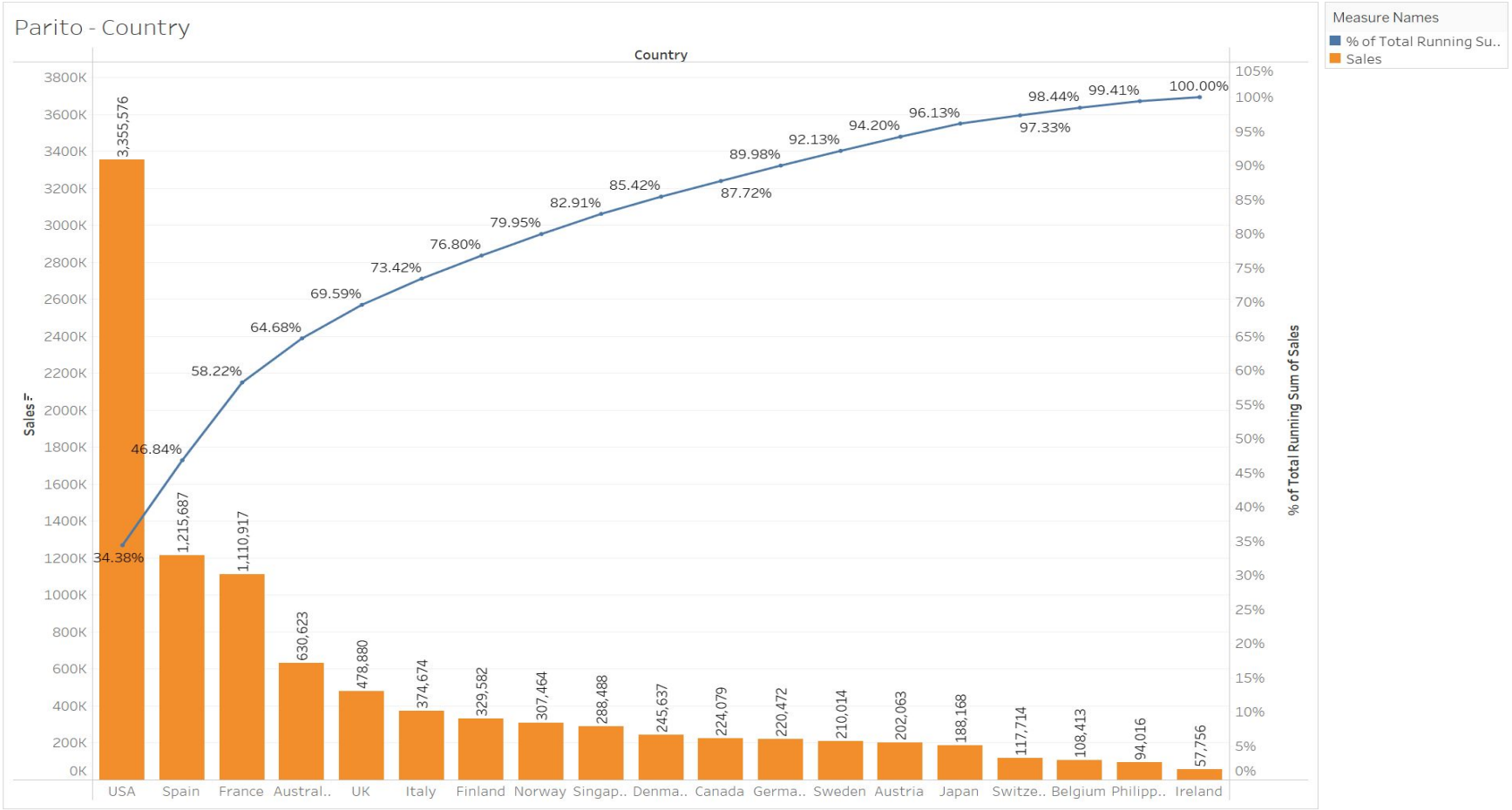
1. Countries with highest sales - value and unit quantity
- **USA ($3.35 Mn - 33,020)**
- **Spain($ 1.21 Mn - 12,429)**
- **France($1.11 Mn - 11,090)**

2. Countries with the lowest sales - value and unit quantity
- **Ireland (0.057 Mn - 490)**
- **Philippines (0.094 Mn - 961)**
- **Belgium (0.108 Mn - 1074)**

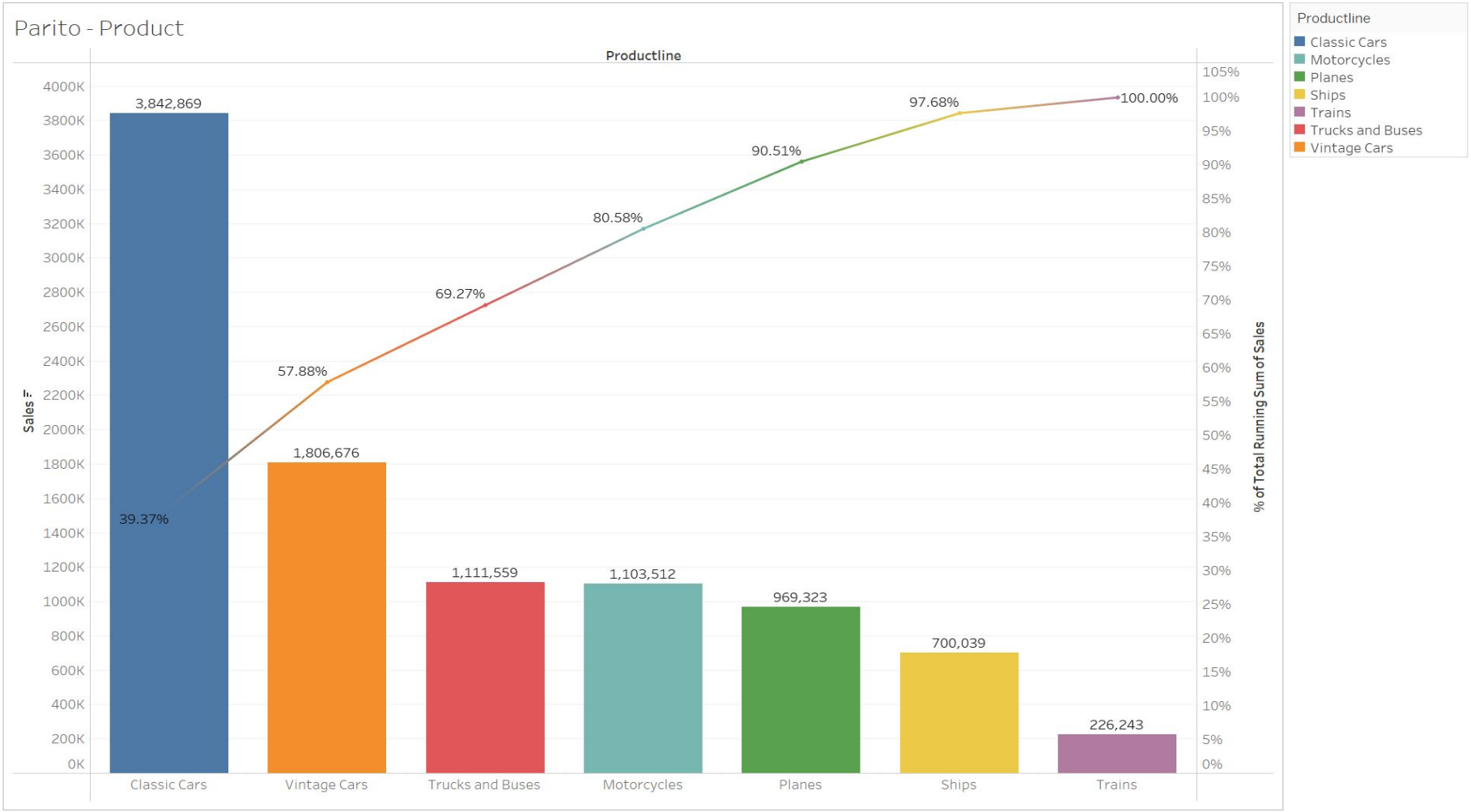
Country wise - Parameter Analysis

Country	Count of Data	Avg. Days Since ..	Avg. Msrp	Avg. Priceeach	Quantityordered	Sales
USA	928	1,737	101	102	33,020	3,355,576
Spain	342	1,860	97	98	12,429	1,215,687
France	314	1,740	100	99	11,090	1,110,917
Australia	185	1,607	100	101	6,246	630,623
UK	144	1,873	96	96	5,013	478,880
Italy	113	1,848	98	100	3,773	374,674
Finland	92	1,752	102	104	3,192	329,582
Norway	85	1,747	105	107	2,842	307,464
Singapore	79	1,750	106	105	2,760	288,488
Denmark	63	1,806	108	110	2,197	245,637
Canada	70	1,923	99	98	2,293	224,079
Germany	62	1,695	101	103	2,148	220,472
Sweden	57	1,566	105	105	2,006	210,014
Austria	55	1,576	103	100	1,974	202,063
Japan	52	1,803	100	101	1,842	188,168
Switzerland	31	1,363	116	112	1,078	117,714
Belgium	33	1,764	101	99	1,074	108,413
Philippines	26	2,080	98	96	961	94,016
Ireland	16	1,939	107	113	490	57,756

-> Almost **58.22%** of sales in 3 countries - **USA, Spain, France**

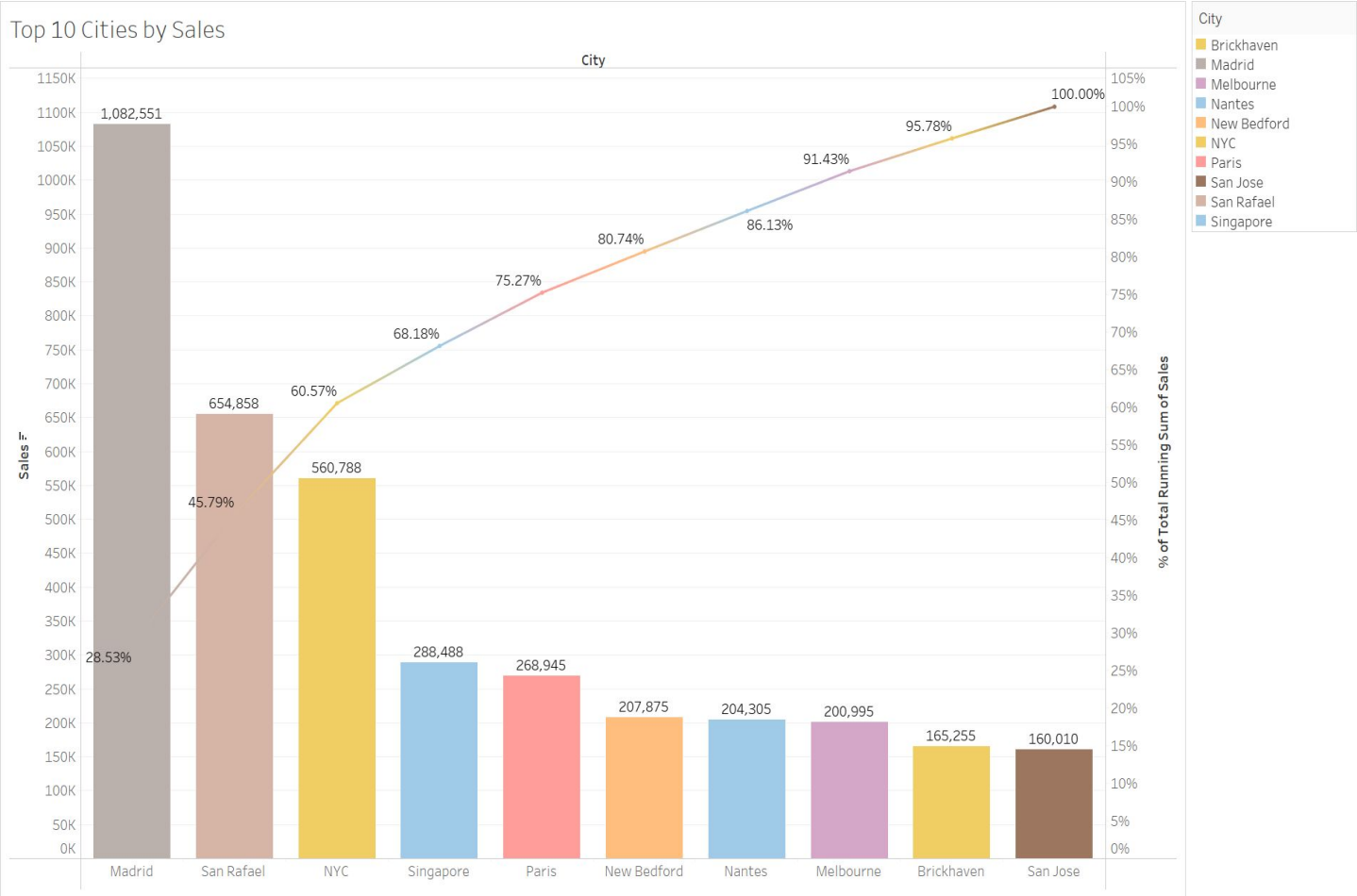


-> **69.27%** of Auto part sales in 3 Product categories - **Classic cars, Vintage cars, Trucks and Buses**



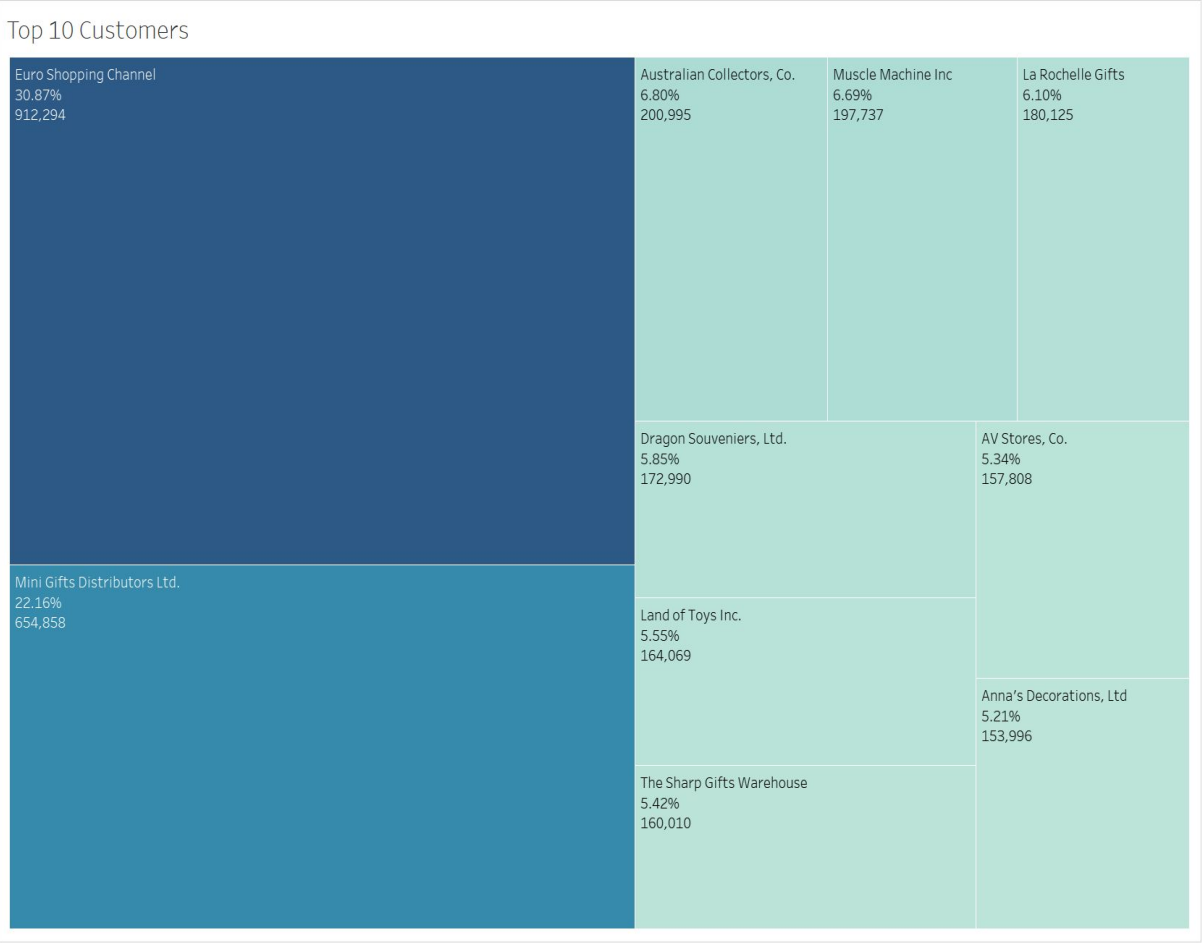
Top 10 Cities by Sales - Auto Parts

- Madrid City has the highest in sales **$1.082** Mn
- San Rafael - **$0.65** Mn
- NYC with **$0.56** Mn

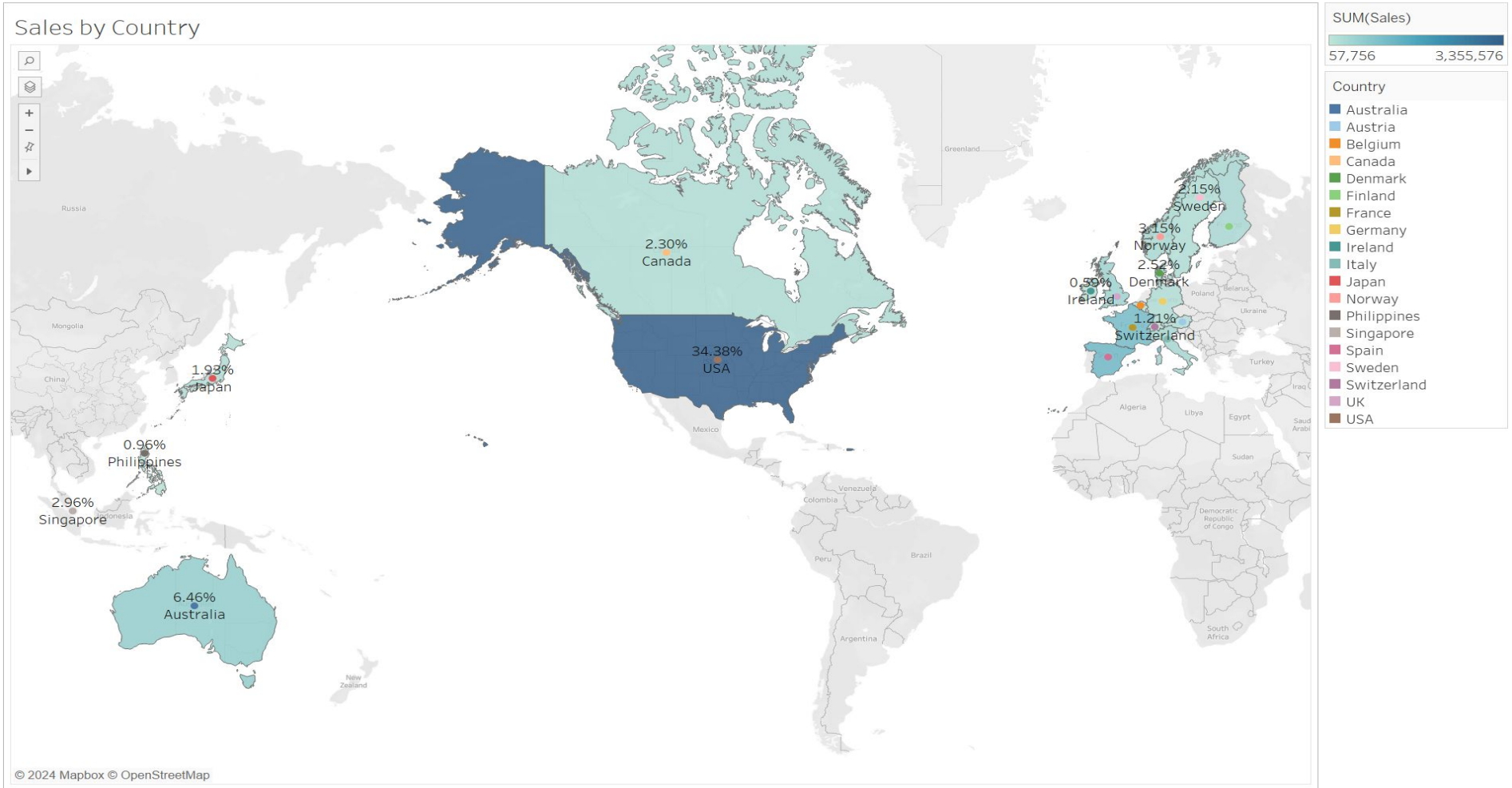


Top 10 Customers by Sales - Auto Parts

- Euro Shopping Channel **30.87 %** with overalls sales (**$0.91** Mn)
- Mini Gifts Distributors Ltd **22.16 %** with overall sales (**$0.65** Mn)
- Australian Collectors, Co 6.8 % with overall sales (**$0.20** Mn)

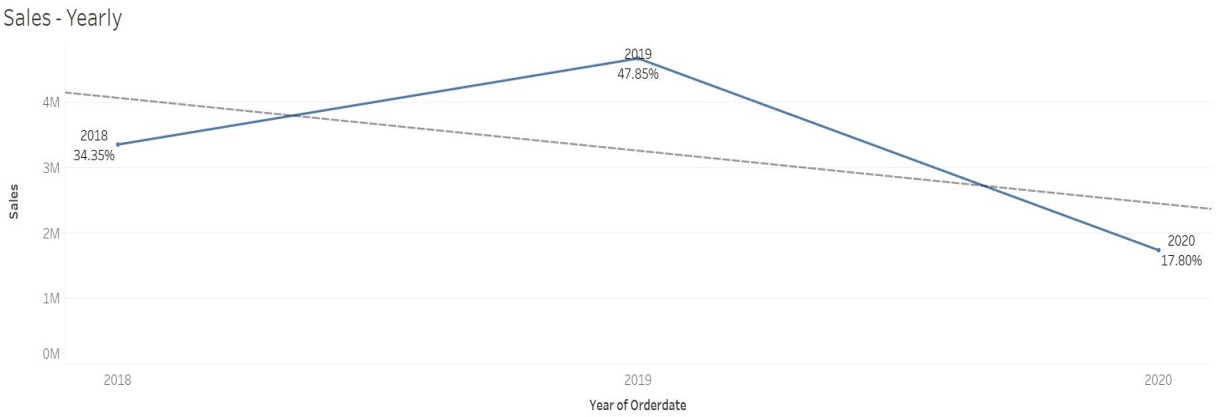


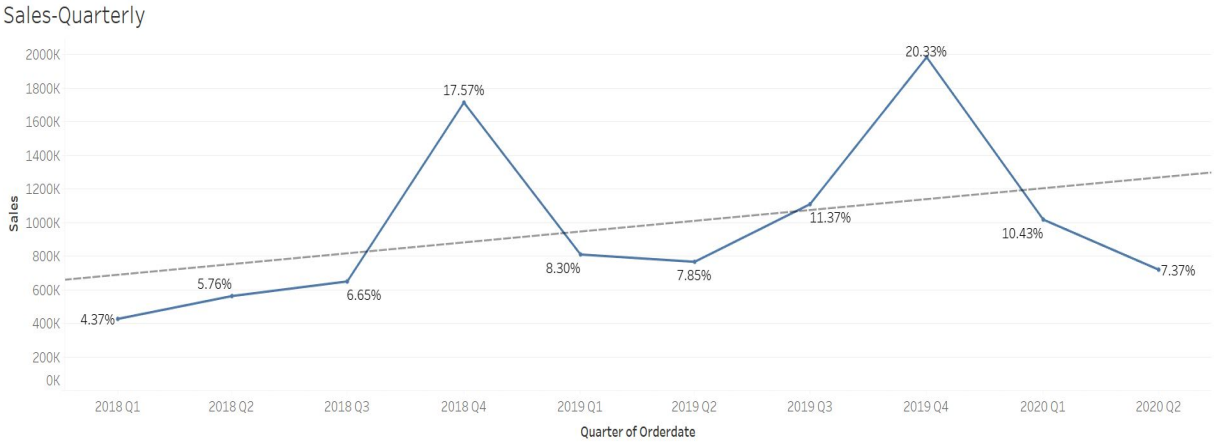
Geographic representation - Auto Parts Sales



Time series Yearly & Quarterly - Auto Parts

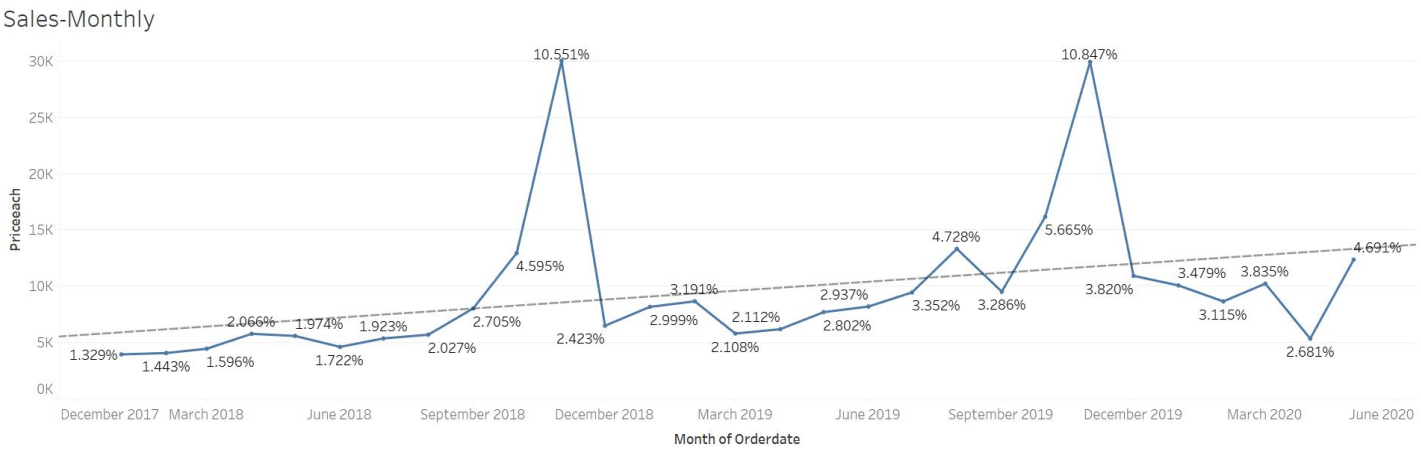
- As per the data, the company done **48%** of sales in 2019.
- Sales was increased in Q4-2018 (**17.57%**), Q4-2019(**20.33%**)

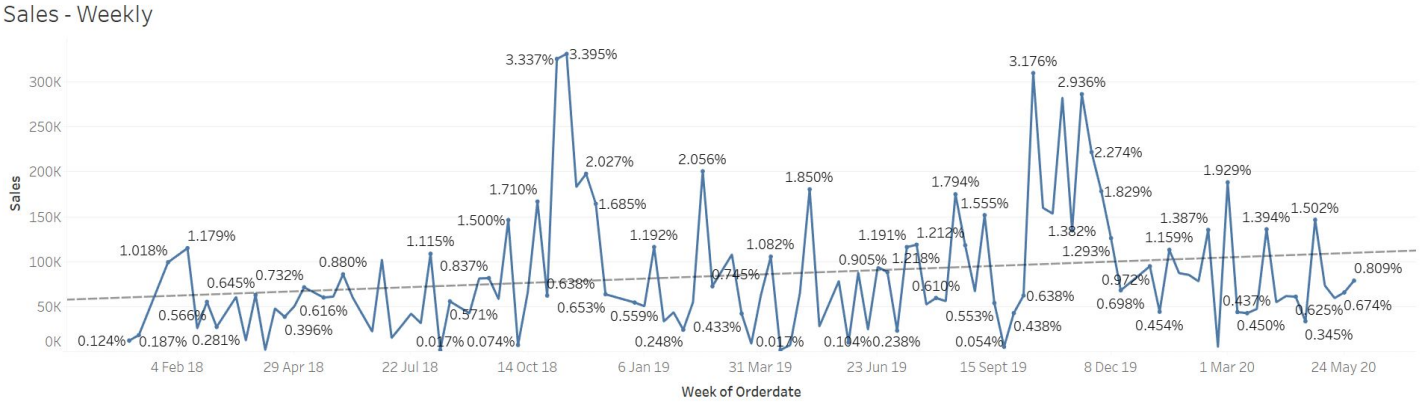




Time series Monthly & Weekly - Auto Parts

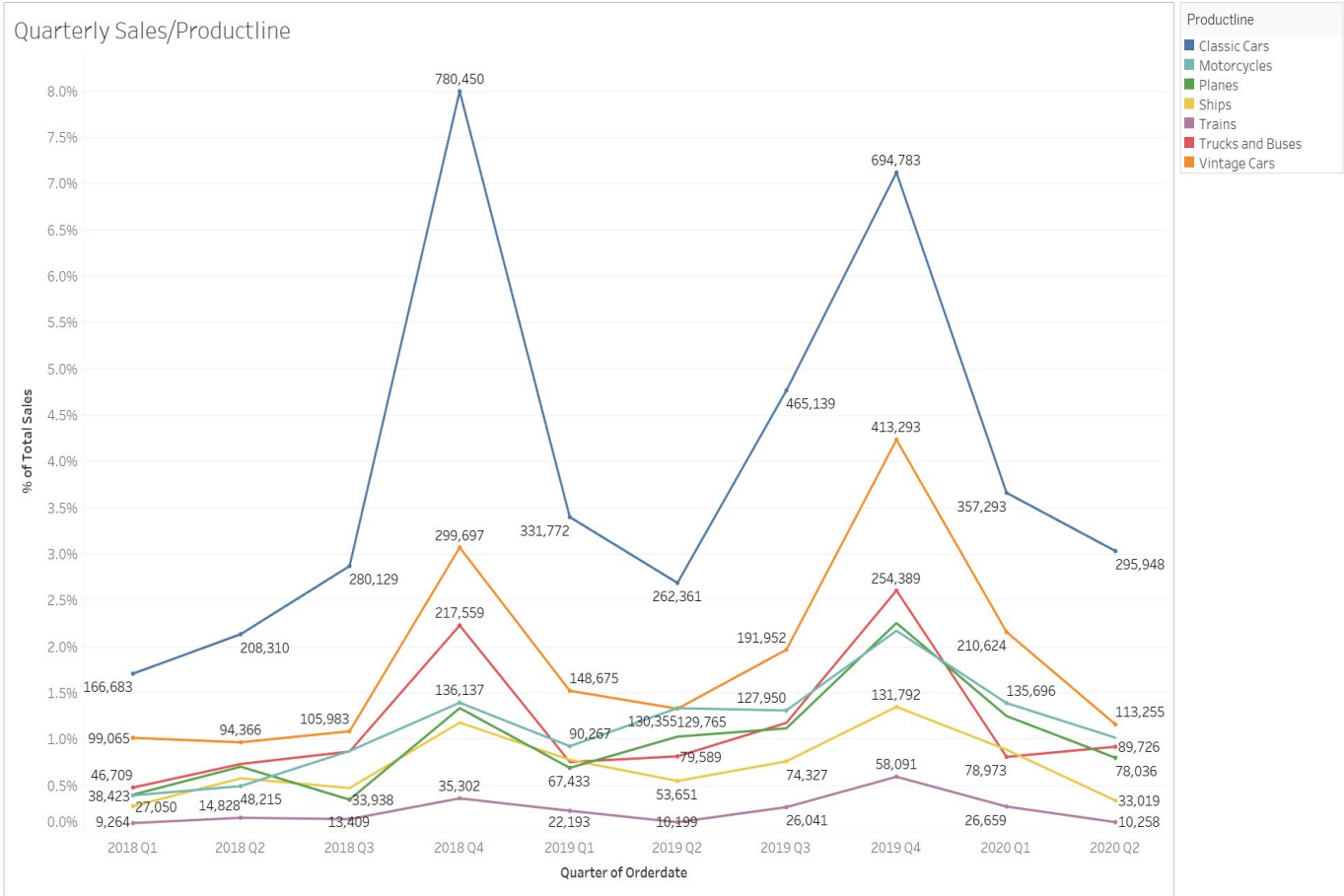
- As per the data, the trendline showing these is a positive sales growth in the Sales.





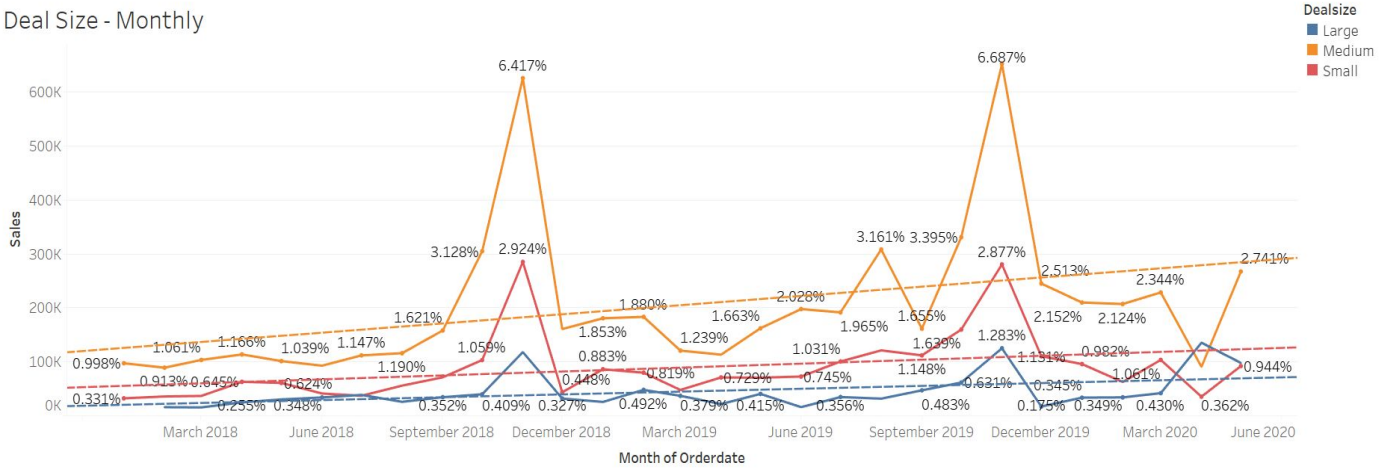
Time series Monthly & Weekly by Product category - Auto Parts

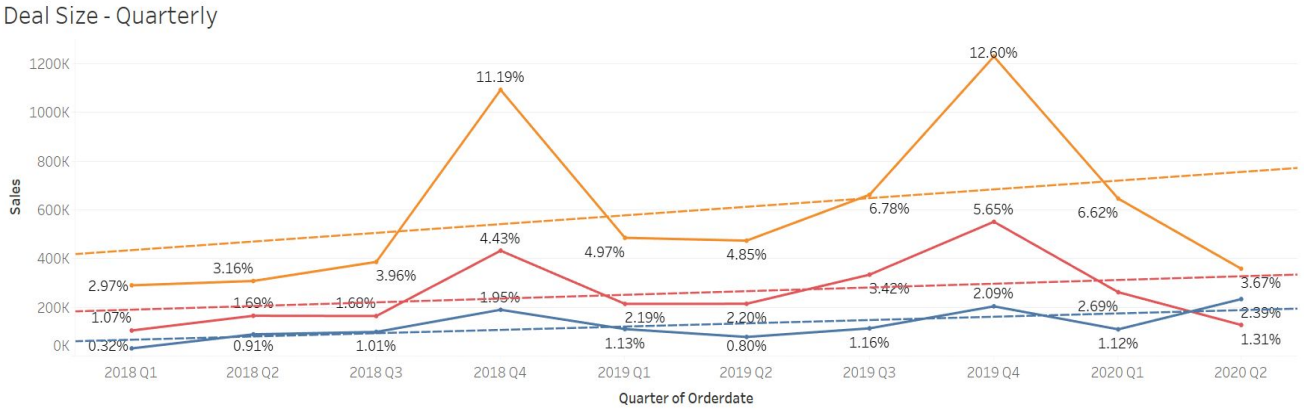
- As per the data, majority of the Auto part orders received for **Classic Cars, Vintage Cars** and **Trucks & Buses.**



Time series Monthly & Quarterly by Deal size - Auto Parts

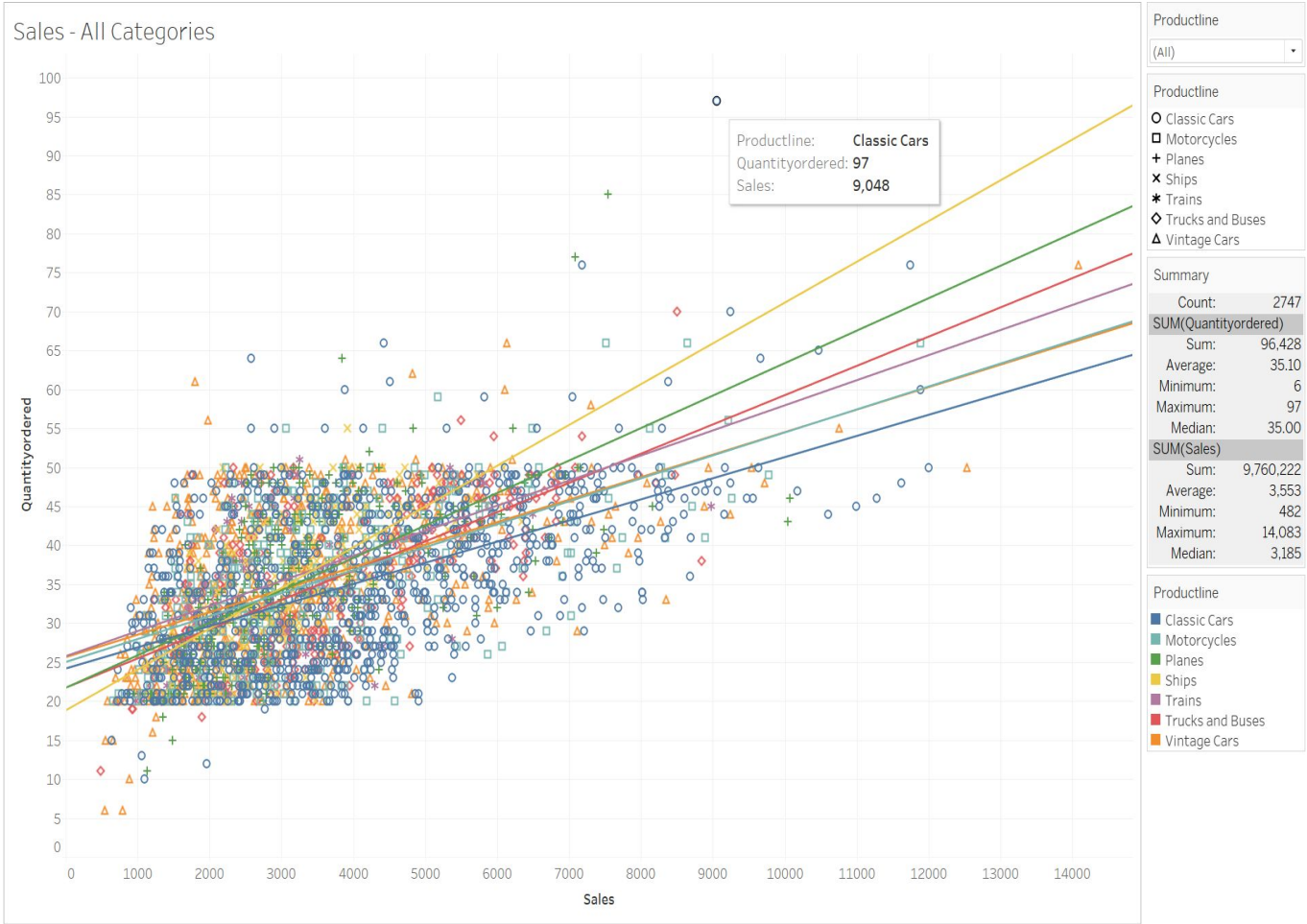
- As per the data, Majority of the orders are in **Medium** by **deal size** in every quarter



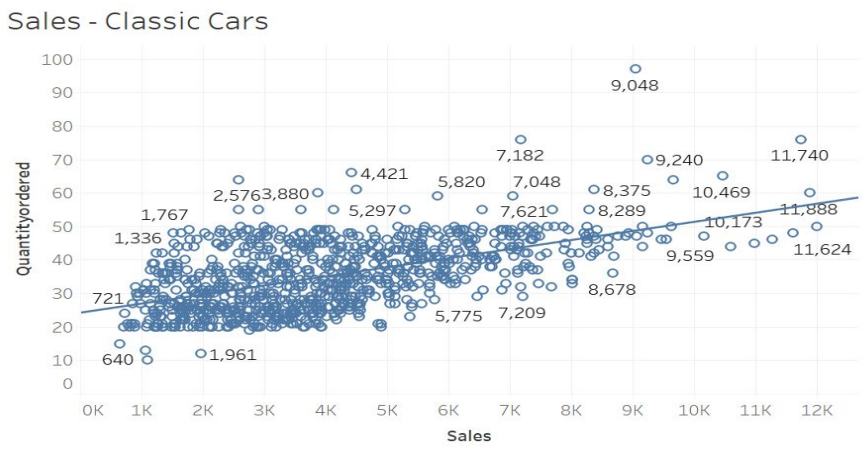


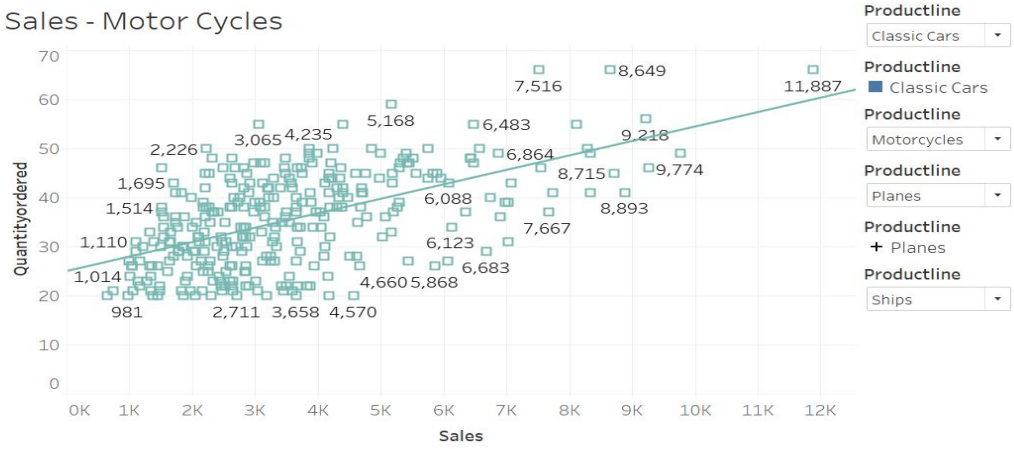
Sales - order quantity by Product Categories- Auto Parts

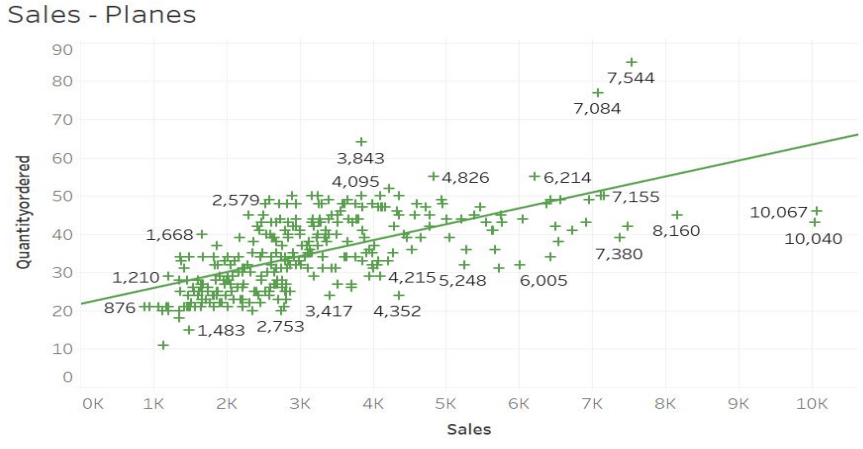
- As per the data, Auto part orders under Classic cars which is Highest **97** orders

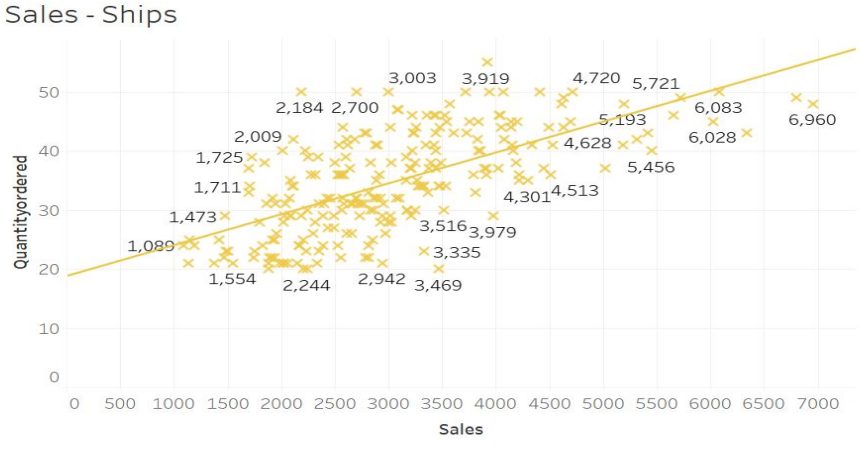


Sales - order quantity for each Product Category- Auto Parts

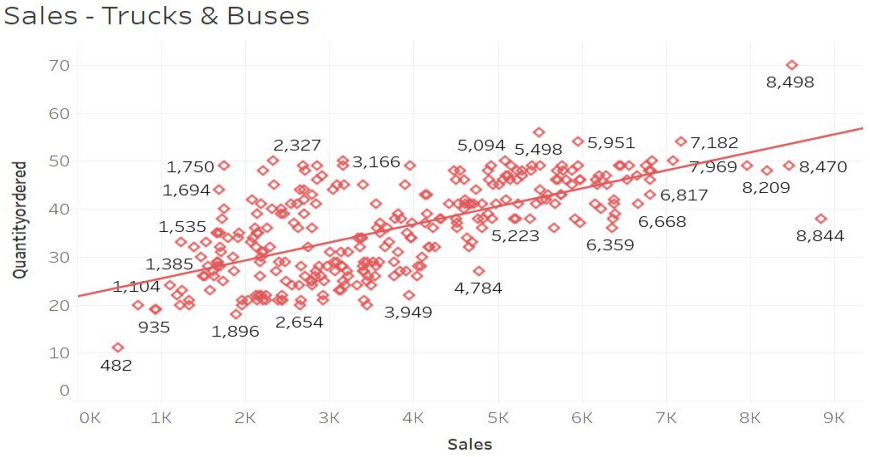


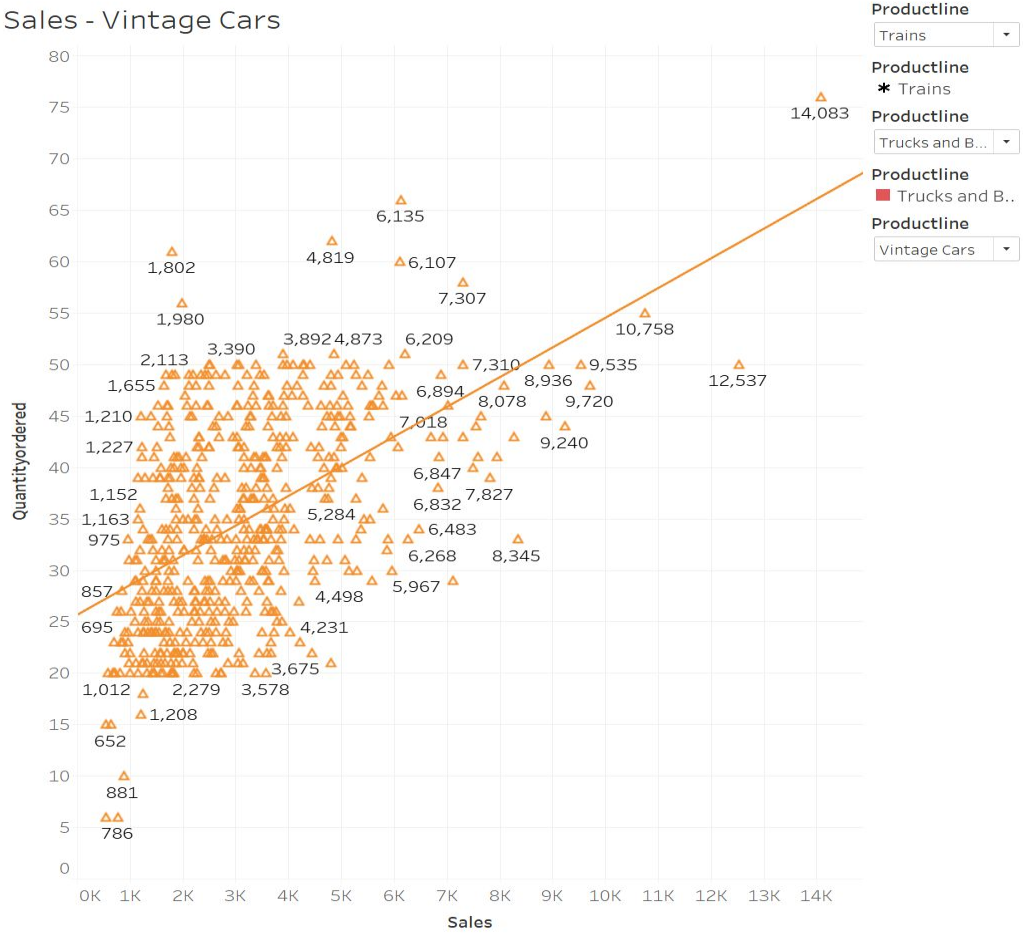






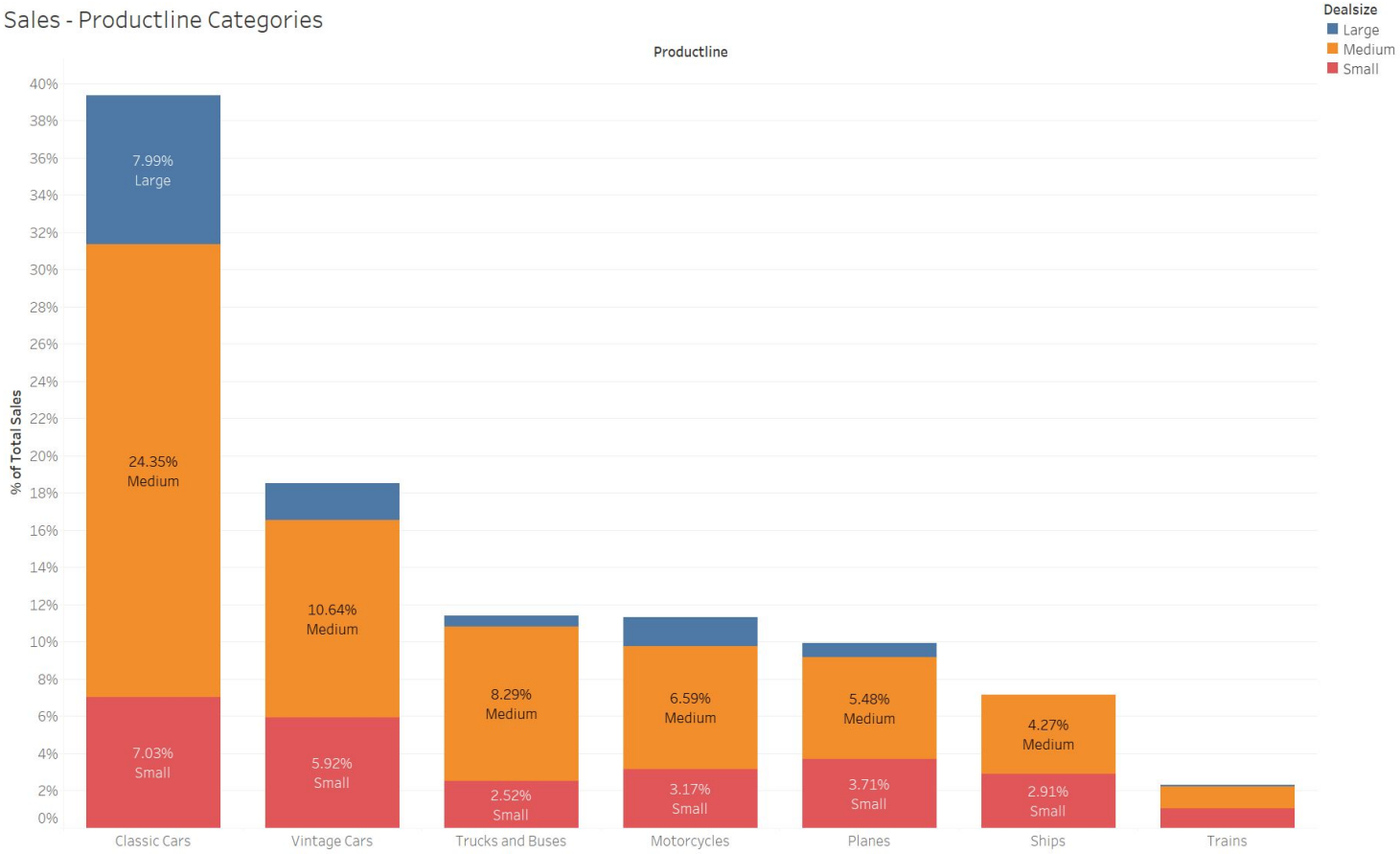
Sales - order quantity for each Product Category- Auto Parts





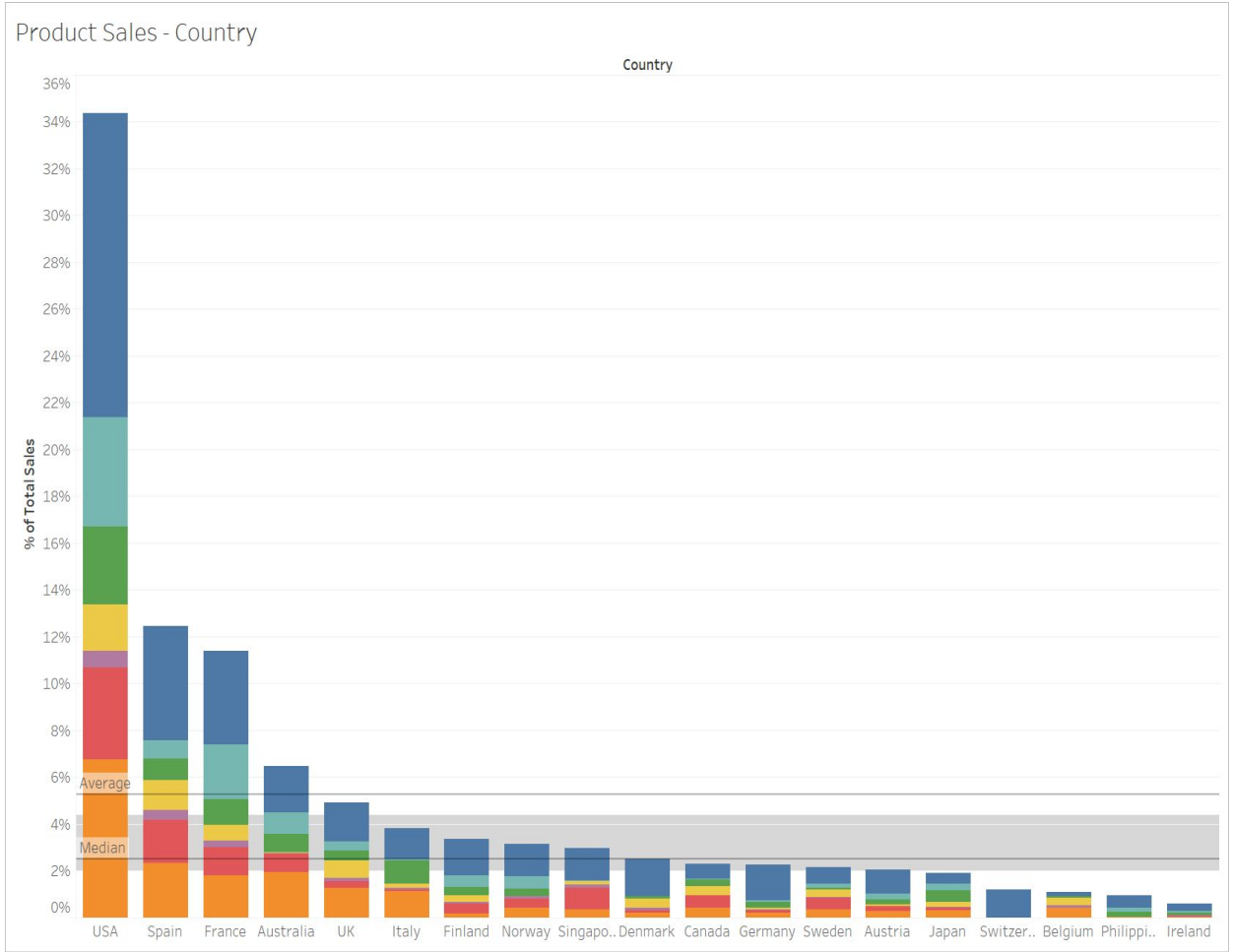
Product categories by Deal size - Auto Parts Sales

- As per the data, Auto part orders larger portions of orders received under **classic cars** with **Medium Deal Size (24.35%)**



Product categories by Country - Auto Parts Sales

- As per the data, Auto part orders larger portions of orders received from the country **USA** markets in which **classic cars** having the highest share in term of sales



Customer Segmentation using RFM analysis

RFM is a customer segmentation technique based on three key metrics:

- Recency (R): How recently a customer made a purchase.
- Frequency (F): How often a customer makes a purchase.
- Monetary Value (M): How much money a customer spends.

RFM Parameters Used:

- Recency (R): **Days since the last order**
- Frequency (F): **Number of orders**.
- Monetary Value (M): **Total sales value.**

Assumptions Made:

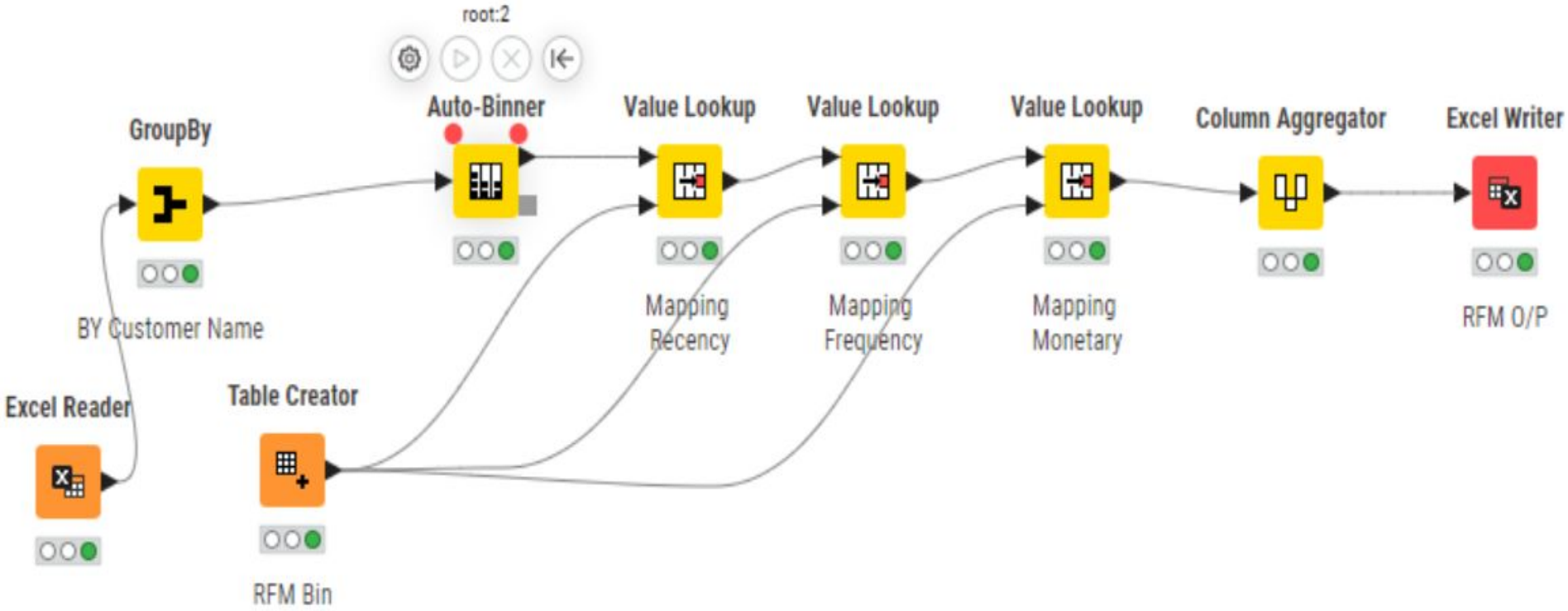
- **Recency (R):** Lower recency values indicate more recent purchases.
- **Frequency (F):** Higher frequency values represent more frequent purchases.
- **Monetary Value (M):** Higher monetary values represent higher spending.

RFM output table head

1	CUSTOMERNAME	Sum(QUAI	Sum(SALES	Mean(DAY	Sum(QUAI	Sum(SALES	Mean(DAY	Recency	Frequency	Monetary	RFM
2	AV Stores, Co.	1778	157807.8	1803.804	Bin 1	Bin 1	Bin 2	3	1	1	311
3	Alpha Cognac	687	70488.44	2236.2	Bin 1	Bin 1	Bin 4	1	1	1	111
4	Amica Models & Co.	843	94117.26	1318.615	Bin 1	Bin 1	Bin 1	4	1	1	411
5	Anna's Decorations, Ltc	1469	153996.1	1463.587	Bin 1	Bin 1	Bin 1	4	1	1	411
6	Atelier graphique	270	24179.96	1424.429	Bin 1	Bin 1	Bin 1	4	1	1	411
7	Australian Collectables,	705	64591.46	1689.696	Bin 1	Bin 1	Bin 2	3	1	1	311
8	Australian Collectors, C	1926	200995.4	1698.782	Bin 1	Bin 1	Bin 2	3	1	1	311
9	Australian Gift Network	545	59469.12	1349.133	Bin 1	Bin 1	Bin 1	4	1	1	411
10	Auto Assoc. & Cie.	637	64834.32	1266.278	Bin 1	Bin 1	Bin 1	4	1	1	411
11	Auto Canal Petit	1001	93170.66	1763.444	Bin 1	Bin 1	Bin 2	3	1	1	311

Knime workflow diagram :

Read data - Group by Customer name followed by Auto binning the selected parameters then mapping with Assigned RFM scores then RFM segmentation saved in Excel Output



Inferences from RFM Analysis:

Best Customers (Top 5):

1. **Euro Shopping Channel (RFM: 344):**
   - High frequency and monetary value
   - Recent purchases
2. **Mini Gifts Distributors Ltd. (RFM: 333):**
   - High frequency, monetary value, and recency
3. **Australian Collectors, Co. (RFM: 311):**
   - Balanced in recency, frequency, and monetary value
4. **La Rochelle Gifts (RFM: 311):**
   - Balanced RFM scores
5. **Land of Toys Inc. (RFM: 311):**
   - Balanced in recency, frequency, and monetary value

Inferences from RFM Analysis:

Customers on the Verge of Churning (Bottom 5):

1. **Giftsbymail.co.uk (RFM: 111):**
   - Low frequency, monetary value, and recency
2. **Quebec Home Shopping Network (RFM: 311):**
   - Balanced but with lower monetary value
3. **Online Diecast Creations Co. (RFM: 411):**
   - Lower recency and frequency
4. **Blauer See Auto, Co. (RFM: 311):**
   - Balanced but lower frequency
5. **Daedalus Designs Imports (RFM: 211):**
   - Lower recency and frequency

Inferences from RFM Analysis:

Lost Customers (Bottom 5):

1. **Giftsbymail.co.uk (RFM: 111):**
   - Lowest RFM scores
2. **Quebec Home Shopping Network (RFM: 311):**
   - Balanced but with lower monetary value
3. **Online Diecast Creations Co. (RFM: 411):**
   - Lower recency and frequency
4. **Blauer See Auto, Co. (RFM: 311):**
   - Balanced but lower frequency
5. **Daedalus Designs Imports (RFM: 211):**
   - Lower recency and frequency

Inferences from RFM Analysis:

Loyal Customers (Top 5):

1. **Euro Shopping Channel (RFM: 344):**
   - High frequency and monetary value
   - Recent purchases
2. **Mini Gifts Distributors Ltd. (RFM: 333):**
   - High frequency, monetary value, and recency
3. **Australian Collectors, Co. (RFM: 311):**
   - Balanced in recency, frequency, and monetary value
4. **La Rochelle Gifts (RFM: 311):**
   - Balanced RFM scores
5. **Land of Toys Inc. (RFM: 311):**
   - Balanced in recency, frequency, and monetary value

Summary and Conclusion:

- The RFM analysis has provided valuable insights into customer segments, allowing for targeted strategies to optimize customer engagement. Understanding the distinct profiles of Best Customers, Customers on the Verge of Churning, Lost Customers, and Loyal Customers guides tailored approaches for retention, re-engagement, and enhancement of loyalty

Best Customers (Top 5):

- Identified customers with high-frequency transactions, significant monetary contributions, and recent interactions. Tailored strategies include maintaining engagement levels, capitalizing on recent interactions for personalized offers, and sustaining loyalty through targeted programs

Customers on the Verge of Churning (Bottom 5):

- Recognized customers displaying signs of potential churn due to lower frequency, monetary value, or recency. Proposed strategies involve implementing re-engagement campaigns, personalized incentives, and retention initiatives to improve RFM scores

Summary and Conclusion:

Lost Customers (Bottom 5):

- Acknowledged customers with the lowest RFM scores or declining engagement. Recovery strategies encompass targeted win-back campaigns, personalized promotions, and addressing concerns to re-establish connections and regain loyalty

Loyal Customers (Top 5):

- Identified customers exhibiting consistent high-frequency transactions, substantial monetary contributions, and recent interactions. Retention strategies focus on acknowledging loyalty through exclusive perks, enhancing VIP programs, and personalizing offerings based on past behaviors

In summary, leveraging RFM analysis empowers businesses to tailor their marketing and retention efforts, fostering stronger connections with customers across various segments. The nuanced understanding of customer behaviors facilitates proactive measures to enhance satisfaction, loyalty, and overall business success